Teacher answers and topic guidance

Accept scientifically equivalent wording and the usual response mode. Keep answers hidden during independent attempts.

Starter

More cells supports division. It does not alone establish differentiation or a specialised role.

Models

The source-range checkpoint accepts embryonic → nerve, muscle or blood; adult blood-forming → blood; meristem → root hair. Differentiation gives a specialised role. Reject animal-to-plant routes and the claim that all adult stem cells form all cell types.

Plant chain: division → elongation → differentiation in this simplified route. The claim about every cell being a stem cell is unsupported by length alone.

Arrival Supported

1. A. Read the label beside the round cell.
2. More cells.
3. The nucleus.
4. Root tip.

Arrival Independent

1. A.
2. More cells with matching inherited information.
3. Division makes more cells; differentiation gives a specialised role.
4. In the root-tip meristem.

Arrival Aspire

1. A, stem cell.
2. Inherited DNA is copied; two daughter cells receive matching inherited information.
3. Division is shown. Differentiation explains specialised roles.
4. The new cells can elongate, increasing their length.

Exit Supported

1. Differentiation.
2. More cells.

Exit Independent

1. Differentiation.
2. Elongation increases cell length; or differentiation gives a specialised role.

Exit Aspire

1. Cell number shows division; a role needs evidence of differentiation and features suited to a function.
2. Elongation or division can contribute; length alone does not identify every cell type or prove stem-cell range.

Topic guidance

Keep stem cell as a cell with a capacity, not a name for every newly made cell. Adult stem-cell types have limited ranges. Embryonic stem cells can form many body-cell types; avoid the claim that all stem cells can form all cell types.

Mitosis is nuclear division. Cell splitting produces separate cells. The teaching drawings simplify the full cell cycle. Mature specialised cells can differ greatly in form; do not claim every specialised cell contains a nucleus.

A feature-function link is an explanation. A schematic drawing supplies a model, not an experimental observation or proof of clinical success.

Made by Matt

Mark scheme Supported

Award each distinct point once. These are question tariffs, not participation scores. Use actual responses and record content support honestly.

1. Finish: Division makes more ___. Differentiation gives a specialised ___. [2]

cells; job.

Maximum 2 marks. Award one mark for each distinct listed science point, up to the tariff. Accept equivalent valid comparisons.

2. Use Reference A. Choose nerve or muscle for a cell with a long extension. Say how that feature helps. [3]

Nerve cell; long extension; carries signals over a distance.

Maximum 3 marks. Award one mark for each distinct listed science point, up to the tariff. Accept equivalent valid comparisons.

3. Choose the possible route: an adult blood-forming stem cell makes blood cells or every kind of body cell. Give a reason. [3]

Blood cells; adult stem-cell types have a limited range; the model does not show every body-cell type.

Maximum 3 marks. Award one mark for each distinct listed science point, up to the tariff. Accept equivalent valid comparisons.

4. Put the plant cards in order: differentiate, divide, elongate. Name the region where new cells are made. [4]

Divide, elongate, differentiate; meristem.

Maximum 4 marks. Award one mark for each distinct listed science point, up to the tariff. Accept equivalent valid comparisons.

5. Compare the skin and root models. Finish: Both make more cells by ___. The root cells also ___. A cell gains a job by ___. [3]

division; elongate; differentiation.

Maximum 3 marks. Award one mark for each distinct listed science point, up to the tariff. Accept equivalent valid comparisons.

Made by Matt

Mark scheme Independent

Award each distinct point once. These are question tariffs, not participation scores. Use actual responses and record content support honestly.

1. Explain the difference between division and differentiation. [2]

Division makes more cells; differentiation makes a cell specialised for a particular job.

Maximum 2 marks. Award one mark for each distinct listed science point, up to the tariff. Accept equivalent valid comparisons.

2. Use a labelled nerve-cell drawing to link one feature to its function. [3]

Labelled long extension; signal transmission; link to carrying a signal over a distance.

Maximum 3 marks. Award one mark for each distinct listed science point, up to the tariff. Accept equivalent valid comparisons.

3. Compare the range of an embryonic stem cell with that of an adult blood-forming stem cell. State one limit of the model. [3]

Embryonic stem cells can give rise to many body-cell types; adult blood-forming stem cells produce a more limited range; model cannot prove successful treatment or show all stages.

Maximum 3 marks. Award one mark for each distinct listed science point, up to the tariff. Accept equivalent valid comparisons.

4. Draw and label a simplified growth route at a root tip. Include division, elongation, differentiation and meristem. [4]

Meristem named; division adds cells; elongation lengthens cells; differentiation gives specialised roles.

Maximum 4 marks. Award one mark for each distinct listed science point, up to the tariff. Accept equivalent valid comparisons.

5. Compare growth and repair in animal skin with growth in a root. Give one shared process and one difference, with an explanation. [3]

Both include division; both may include differentiation; root-tip meristem followed by elongation is a relevant plant difference. Credit an equivalent clear three-point comparison.

Maximum 3 marks. Award one mark for each distinct listed science point, up to the tariff. Accept equivalent valid comparisons.

Made by Matt

Mark scheme Aspire

Award each distinct point once. These are question tariffs, not participation scores. Use actual responses and record content support honestly.

1. A model shows two new cells, both labelled specialised. Explain why cell number alone cannot justify those labels. Add the missing process. [3]

Number supports division; number alone does not show a specialised feature or role; add differentiation with a feature-function link.

Maximum 3 marks. Award one mark for each distinct listed science point, up to the tariff. Accept equivalent valid comparisons.

2. Compare a nerve cell and muscle cell. Link one feature of each to its job, then state what a drawing cannot establish. [4]

Nerve extension; signal transmission; contractile muscle structures linked to contraction; drawing alone cannot establish measured function.

Maximum 4 marks. Award one mark for each distinct listed science point, up to the tariff. Accept equivalent valid comparisons.

3. Evaluate: Every adult stem cell can become any body cell. Correct the scope and use the blood-forming example. [4]

Reject universal claim; adult stem-cell types have limited ranges; blood-forming stem cells make blood cells; embryonic stem cells have a wider range of body-cell types.

Maximum 4 marks. Award one mark for each distinct listed science point, up to the tariff. Accept equivalent valid comparisons.

4. Compare the skin and root routes. Explain two shared ideas and one plant difference. Give one limit of a simple three-step sequence. [5]

Division shared; differentiation shared; root-tip meristem; elongation contributes to length; sequence is simplified and processes can overlap across cells.

Maximum 5 marks. Award one mark for each distinct listed science point, up to the tariff. Accept equivalent valid comparisons.

5. Repair: A longer root proves every new cell is a stem cell. State what growth could show, what it does not show and a relevant extra check. [4]

Length indicates growth; could involve division and elongation; does not establish every cell is a stem cell; inspect cell features or location using an appropriate prepared reference, without claiming that appearance alone proves potency.

Maximum 4 marks. Award one mark for each distinct listed science point, up to the tariff. Accept equivalent valid comparisons.

Made by Matt

Mark scheme Choice

Award each distinct point once. These are question tariffs, not participation scores. Use actual responses and record content support honestly.

1. Choose division or differentiation for makes more cells. Choose again for gives a job. Add one short reason. [2]

division; differentiation, with a matching reason.

Maximum 2 marks. Award one mark for each distinct listed science point, up to the tariff. Accept equivalent valid comparisons.

2. Choose nerve or muscle for carrying signals. Choose long extension or no structure as a useful feature. Explain one link. [3]

nerve; long extension; carries signals over a distance.

Maximum 3 marks. Award one mark for each distinct listed science point, up to the tariff. Accept equivalent valid comparisons.

3. Choose blood cells or all body cells for the range shown from an adult blood-forming stem cell. Say one limit. [3]

blood cells; limited adult range; does not prove treatment success.

Maximum 3 marks. Award one mark for each distinct listed science point, up to the tariff. Accept equivalent valid comparisons.

4. Select a plant sequence: divide then elongate then differentiate, or differentiate then no further change. Name root-tip meristem or leaf middle as the dividing region. [4]

divide, elongate, differentiate; root-tip meristem.

Maximum 4 marks. Award one mark for each distinct listed science point, up to the tariff. Accept equivalent valid comparisons.

5. Select both, animal only or plant only for division. Select elongation or signal transmission as the length change in the root model. Give one comparison sentence. [3]

both; elongation; a clear comparison linking the process to its effect.

Maximum 3 marks. Award one mark for each distinct listed science point, up to the tariff. Accept equivalent valid comparisons.

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